🛠 Engineering Team Charter

fs881, bta26, ac3847

Team Name:

Home Audio System Designers

Mission Statement:

Our mission is to collaboratively design, build, and verify a complete home audio system, integrating analogue circuit design, Class D switching amplifier electronics, and Arduino embedded programming into a fully functional prototype.

Through clear role allocation, daily communication, and systematic bench verification at every stage, we aim to deliver a system that meets all design specifications and demonstrates sound engineering practice within the five-day project week.

Goals & Objectives:

- Goal 1: On Day 1, assign lead responsibilities for circuit design, embedded programming, and bench testing. Establish the shared project folder, agree the file naming convention, and create the file index before any practical work begins. Set up a daily morning check-in to review progress and reprioritise tasks.
- Goal 2: On Day 2, select component values for all three MFB bandpass filters and verify they meet the frequency and gain targets. Build and verify the full graphic equalizer on the breadboard using the function generator and oscilloscope, confirming neutral response, 12 dB maximum gain per band, and ripple below 4 dB. Implement and verify the OLED welcome page.
- Goal 3: On Day 3, build the triangle wave integrator and verify the 100 kHz waveform on the oscilloscope. Build the LM393 comparator and verify the PWM output. Build the half-bridge switching stage with dead time circuit and confirm dead time is equal on both transitions with no shoot-through current.
- Goal 4: On Day 4, measure earphone DC resistance with the DMM, select and build the matched RLC output filter, and confirm ripple below 100 mV and DC offset below 0.4 V. Build peak detector hardware for all three bands, wire to the Arduino analog inputs, and complete the spectrogram code.
- Goal 5: On Day 5, connect all three subsystems and verify the complete audio chain from input to earphone output. Confirm the OLED spectrogram responds smoothly to potentiometer adjustment across all three bands. Complete the QMS check: verify the file index is complete and all milestone evidence is saved.

Team Members & Roles:

Name	Role	Responsibilities
bta26	Project Lead and Circuit Design Lead	Responsible for overall project coordination, graphic equalizer circuit design and component selection, Design Verification oversight, and ensuring the team is on track at each daily check-in.
fs881	Embedded Systems Lead	Responsible for all Arduino firmware including PWM configuration, OLED welcome page, and spectrogram code. Also leads technical documentation of code and embedded design decisions.
ac3847	Amplifier and Testing Lead	Responsible for the Class D modulator, switching stage, and RLC filter construction. Leads all oscilloscope and DMM bench measurements and confirms each subsystem passes its verification criteria before integration.

Although lead responsibilities are assigned, the project is built collaboratively. Team members support each other during breadboarding and bench work, and this cooperative approach ensures knowledge is shared across all three subsystems.

Resources & Constraints:

- Tools: Arduino IDE for all embedded programming. A shared project folder with an agreed file naming convention is used for version control and evidence storage throughout the week.
- Equipment and Materials: Two breadboards, bench DC power supply (+-5 V), oscilloscope, function generator, and DMM. Arduino Nano, SSD1306 OLED display, 3.5 mm earphones, and standard electronic components including TL071 op-amps, LM393 comparator, BS250 PMOS, 2N7000 NMOS, and BC547 NPN transistors from the laboratory store.
- Budget: No direct financial budget. All components are provided through the university laboratory store. Component availability acts as a practical constraint and designs use only E12 standard resistor values to ensure all required parts are accessible.
- Constraints: The project must be completed within five lab days. All measurements use the oscilloscope, function generator, and DMM. Earphones must not be connected until DC offset is confirmed below 0.4 V. Only an Arduino was available to be used with Wire.h, Adafruit_SSD1306.h, and Adafruit_GFX.h libraries.

Operating Norms:

Build-and-test discipline:
The team follows a staged build process where each circuit block is first checked in isolation, then connected to the next stage only after its input, output, supply rails, and expected signal behaviour have been confirmed. This reduces fault propagation and makes integration easier to debug.

Bench setup control:
At the start of each practical session, the team confirms that the oscilloscope probes, function generator settings, power supply limits, and ground connections are correct before energising any circuit. This prevents avoidable damage from incorrect supply polarity, excessive signal amplitude, or floating references.

Change tracking:
Any component value change, wiring change, or code update is recorded immediately in the shared notes before further testing. This ensures the team can trace what was changed if the system behaviour improves, worsens, or becomes inconsistent.

Evidence capture during testing:
Test evidence is captured at the moment a result is achieved, not later from memory. Screenshots must show useful scale settings where possible, and photographs should clearly show the circuit section being tested. This avoids losing key evidence needed for Mahara.

Debugging approach:
When a fault occurs, the team works backwards from the output to the input using measured signals rather than randomly replacing components. Only one change is made at a time so the cause of improvement or failure can be identified clearly.

End-of-session handover:
Before leaving the lab, the team spends a few minutes noting what currently works, what still fails, what was last changed, and what the next practical step should be. This allows the next session to begin efficiently without repeating previous debugging.

Decision-Making Process:

Specification-led decisions:
The team makes design choices based on whether they help meet the stated project requirements, such as frequency response, gain, ripple, DC offset, dead time, and OLED functionality. A solution that performs reliably is prioritised over one that is more complex but harder to verify.

Measurement-based comparison:
When two possible approaches are available, the team compares them using quick calculations, simulation results, or bench measurements. The option with clearer evidence and lower integration risk is selected.

Subsystem ownership with team review:
Each technical lead proposes decisions within their area, but changes that affect another subsystem are reviewed by the whole team before implementation. For example, changes to the equalizer output level must be checked against the Arduino input and amplifier stage.

Risk-based prioritisation:
If time becomes limited, decisions are prioritised by project risk. Safety-critical and integration-critical tasks, such as DC offset checks, power stage verification, and complete audio-chain testing, take priority over cosmetic improvements.

Escalation route:
If the team cannot agree on a technical direction, the issue is reduced to a specific testable question. The team then uses a measurement, datasheet limit, or supervisor guidance to make the final decision.

Conflict Resolution:

Step 1: Clarify the disagreement:
The team first defines exactly what is being disputed, such as a component value, test method, wiring approach, or interpretation of a result. This prevents the discussion becoming vague or personal.

Step 2: Refer back to requirements:
The disagreement is compared against the project specification and verification targets. The option that better satisfies the measurable requirement is preferred.

Step 3: Test instead of debate:
Where possible, the team resolves technical disagreements by testing both options quickly on the bench or in simulation. The result with stronger measured performance is selected.

Step 4: Protect project progress:
If a disagreement is taking too long, the team chooses the lower-risk option that allows progress to continue and records the decision in the project notes. The alternative can be revisited only if time remains.

Step 5: Handle interpersonal issues separately:
If the conflict is about communication, workload, or team behaviour rather than technical content, it is discussed calmly and directly. If it begins to affect the project output, the issue is raised with the supervisor.

Team Values & Culture:

• **Safe testing first:** The team checks supply voltages, grounding, current limits, and DC offset before connecting sensitive components such as earphones or the Arduino input.

• **Traceable engineering decisions:** Circuit changes are not treated as guesses. Each important decision should be linked to a calculation, simulation, datasheet, or measurement.

• **One-system mindset:** Although each member leads a different area, the final product is judged as one integrated system. Every member is expected to understand the interfaces between subsystems.

• **Efficient debugging:** The team avoids changing multiple things at once. Fault-finding should be logical, measured, and documented so that successful fixes are repeatable.

• **Practical realism:** The team favours solutions that can be built, tested, and evidenced within the five-day project week rather than overcomplicated ideas that increase risk.

• **Accountability without blame:** If something fails, the focus is on identifying the cause and fixing it, not blaming the person who built or tested that section.

Sign-Off & Commitment:

Freddy Sayegh

Ayush Chauhan

Benjamin Annetts